CAN connection and SocketCAN setup

Developer guide V0.1 | Source software v1.1.0 | 2026-10-11

Luchs A connection reference and conditional B interface setup. Confirm the delivered harness before wiring.

Status: Software reference

What this connection does

CAN is the wired link between the computer and chassis. SocketCAN makes a supported Linux CAN adapter available as an interface such as can0. CAN carries commands and feedback; ROS sits above that connection.

Chassis/interface	Public source reference	Required check
Luchs A v1.1	1 Mbit/s; standard 11-bit frames	Delivered connector and firmware
B draft LCI	Separate customer CAN; 1 Mbit/s	LCI availability and released harness
B VCU V2.6 / legacy_b	500 kbit/s is a source expectation	Confirm bitrate and ID numbering on this unit

Scope note. Do not apply the LCI setup to a VCU bus. The B connector and pinout remain unit-specific in the public source; request the matching harness information before connecting.

A signal expansion connector

The public A source locates the signal expansion connector under the magnetic cover at the rear of the deck. Check the delivered cable against this reference:

Pin	Wire colour	Signal
1	Black	RS-232 RX
2	Brown	RS-232 TX
3	Red	GND
4	Orange	CAN-H
5	Yellow	CAN-L
6	Green	RS-485 A
7	Blue	RS-485 B
8	Grey	GND
9	White	GND

Use CAN-H, CAN-L and the signal ground for the CAN adapter. RS-232 and RS-485 are separate signals on the connector. The public source calls for 120 ohm termination at both ends; check the existing termination before enabling an adapter terminator.

Bring up a confirmed A interface

1. Connect a Linux-supported USB-CAN adapter and inspect its interface name.

2. With the vehicle secured against unintended movement, configure the confirmed A bus bitrate.

3. Inspect interface status and listen for feedback before launching the driver.

```
ip -br link
sudo ip link set can0 down
sudo ip link set can0 type can bitrate 1000000 restart-ms 100
sudo ip link set can0 txqueuelen 1000
sudo ip link set can0 up
ip -details -statistics link show can0
candump can0
```

candump is supplied by the Linux can-utils package. The existing software archive also provides luchs_bringup/scripts/setup_can.sh; run it with sudo after reviewing the interface name and confirmed bitrate.

A feedback references include 0x5B5 through 0x5BC. No traffic: check chassis power, adapter detection, CAN-H/L, ground, termination and bitrate. ERROR-PASSIVE or BUS-OFF indicates a bus problem; resolve it before sending commands.

B connection boundary

LCI and legacy_b have different message layouts and different bitrate assumptions. Use passive capture and the unit documentation to identify the interface. Do not choose 500 kbit/s, 1 Mbit/s or a frame ID solely because it is a driver default.

A parameter or harness change must be recorded with the unit and firmware version. This guide supplies no B pin assignment, firmware image or maintenance reset procedure.

Source and edition

Prepared and reorganised by RJ Tech from the already-public Luchs ROS 2 v1.1.0 package by TerraAgent Robotics. Source licence: Apache-2.0. Source text and examples were shortened, explained and scoped for this topic. Original archives and protocol files remain unchanged.

Source file: docs/can_setup.md

Source file: luchs_bringup/scripts/setup_can.sh

Original public software archive | Source licence